

GUARDIANS OF THE GRAPEVINE

“Saving your harvest, saving our wine”

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MAE 2250 | Cornell University | Duffield College of Engineering



1. PROBLEM & WHY IT MATTERS

An invasive species to the United States, Spotted Lantern Flies (SLF) caused \$8.8 Million of losses to NY alone in 2025. They feed on grapevines, damaging the plant and reducing wine yield. They also contaminate the harvest, and hence devastate the Wine Industry.



Fig. 1: A Spotted Lanternfly. We can see its unique pattern.



Fig. 2: Many SLFs on a vine. We can see how they are easy to spot, come in great numbers and can swarm grapevines causing extreme damage.

2. SOLUTION DIRECTION

Develop a non-chemical, selective and scalable device that attracts and eliminates SLFs to protect the grapevines and preserve harvest quality.

Our basic concept was a pole vibrating at 60 Hz (research showed that SLFs are attracted to that frequency). We would then electrocute the pole to eliminate the SLFs

3. FIRST PROTOTYPE

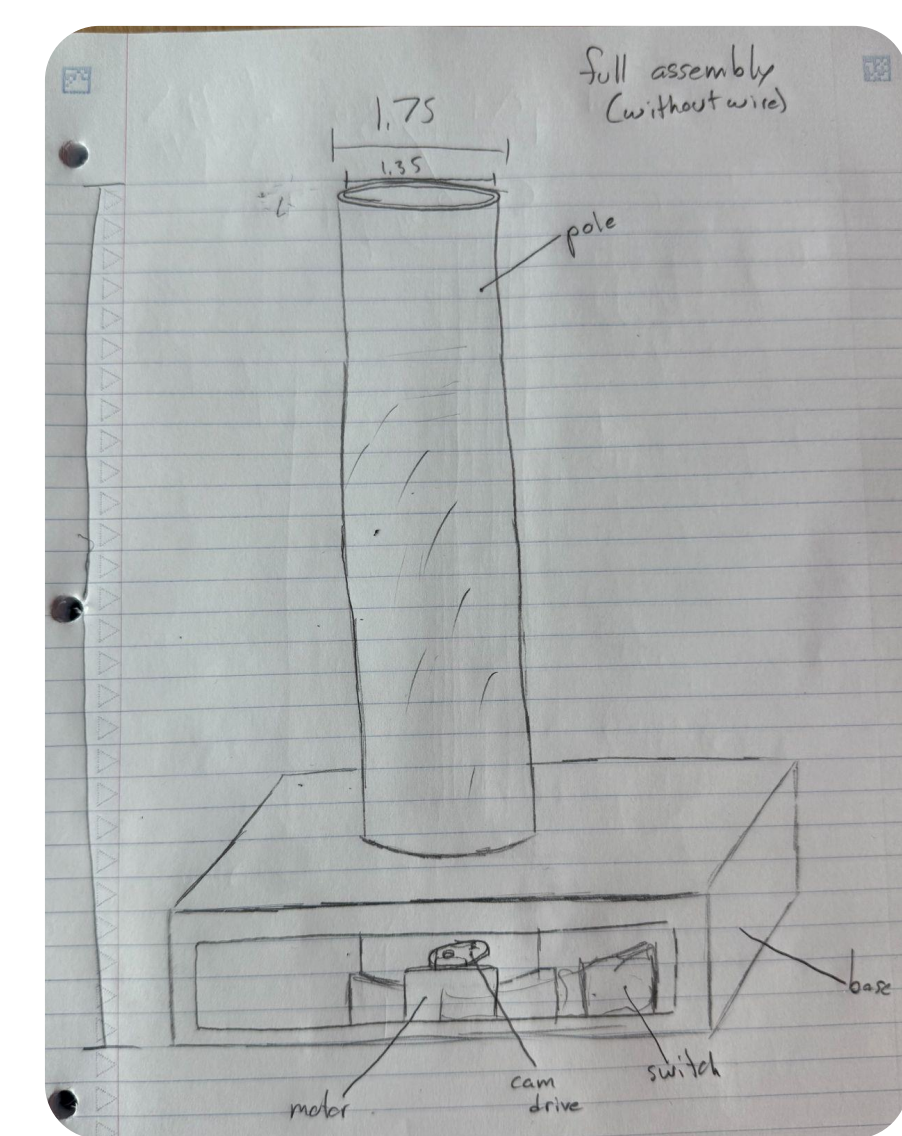


Fig. 3: Sketch Of Our First Prototype

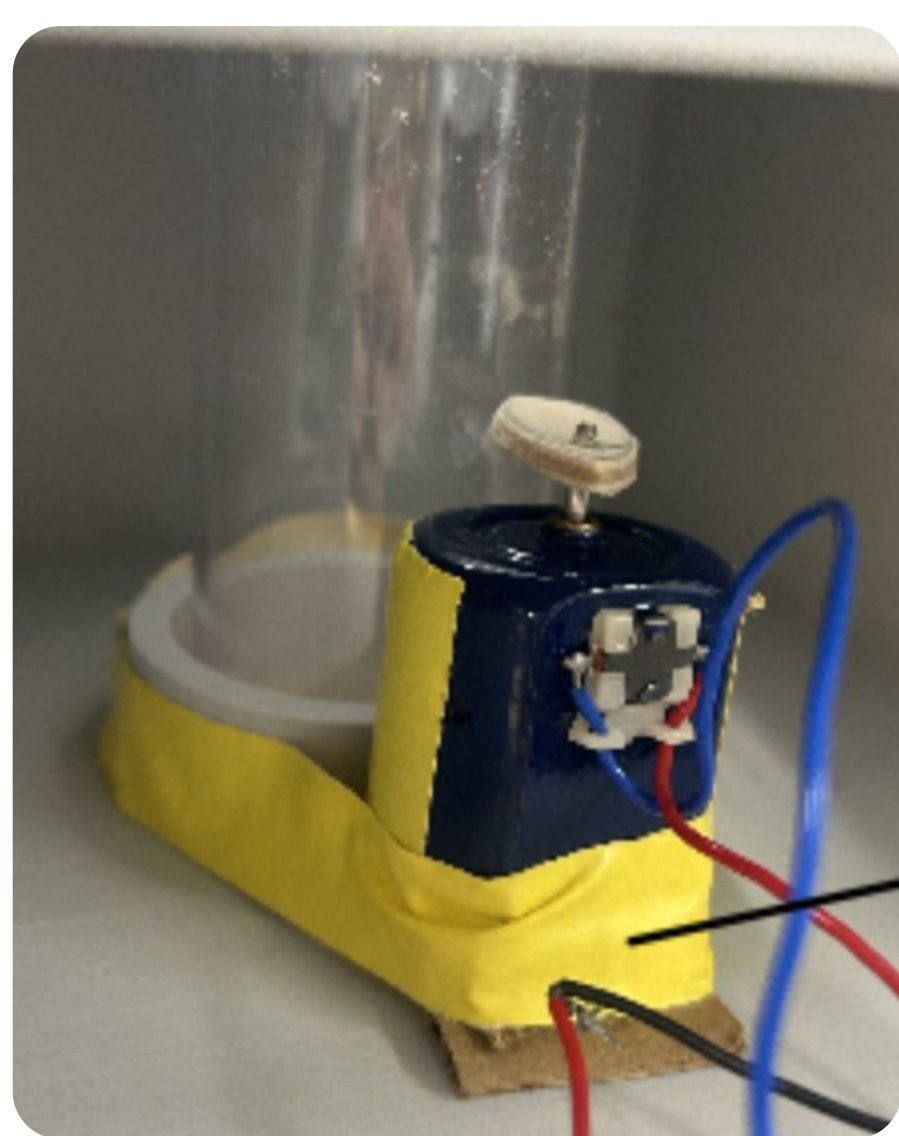


Fig. 4: The Vibration Mechanism

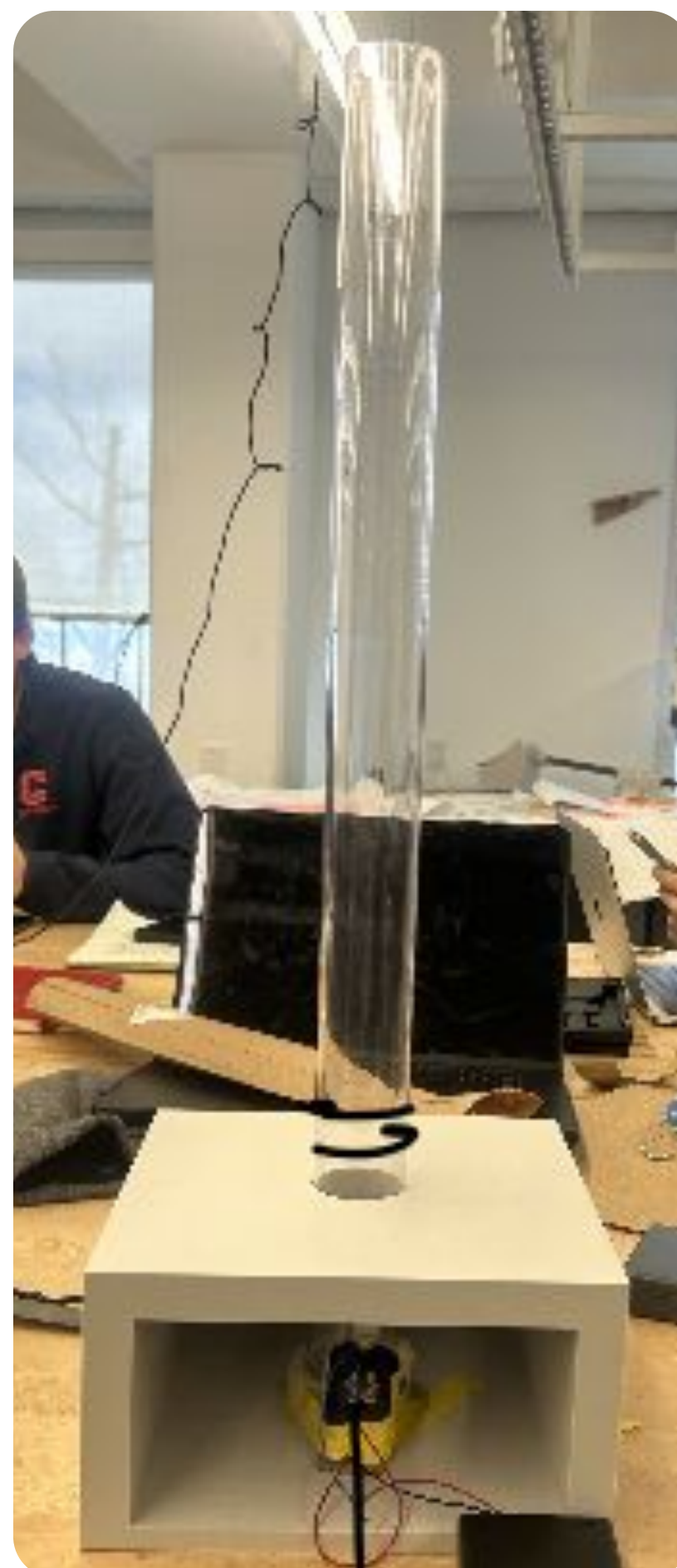


Fig. 5: Our First Prototype

Fig. 3:

Here we realised the main components would be the vibrating tube, the base plate and the vibrating mechanism. The base plate would house all of the mechanical components.

Fig. 4:

We taped a motor with a rotating cam to the pole to induce vibrations via the cam 'hitting' the pole.

Fig. 5:

Notice the pole extrudes from the baseplate and all of the "innards" are housed in the base plate

4. DESIGN PROCESS TIMELINE

1

Research & Problem Definition

2

Concept Development & CAD

3

Prototype Build

4

Testing & Validation

5

Final Design & Refinement

5. SUCCESS CRITERIA

To ensure our project worked to our standards, we devised a series of success criteria to gauge the machines functions.

1

Vibration Frequency

We need to ensure the pole vibrates at 60Hz so that SLFs are attracted to the pole

2

Structural Stability

We want to ensure the structure can support the weight of many SLFs

3

Continuous Vibrations

We want to ensure the mechanism can operate for long periods of time without human interaction

6. TESTING AND IMPROVEMENTS

- PhyPhox (an app on a smartphone) was used to measure the frequency. It was 25 Hz.

Improvement: using gear ratios to have the driven shaft spin at 3600rpm (equivalent of 60Hz)

- To test continuity, we ran the machine for 5 minutes. The mechanism was still in place and running but the striking wooden cam caused damage to the pole.

Improvement: change the wooden cam to a softer rubber cam that would spin inside of the tube

- We applied a lateral force to the pole, reaching 5N without deformation. For the prototype, we were happy with but for final product we wanted to improve
- Improvement:** making the hole in the base plate slightly narrower so that pole had less slip

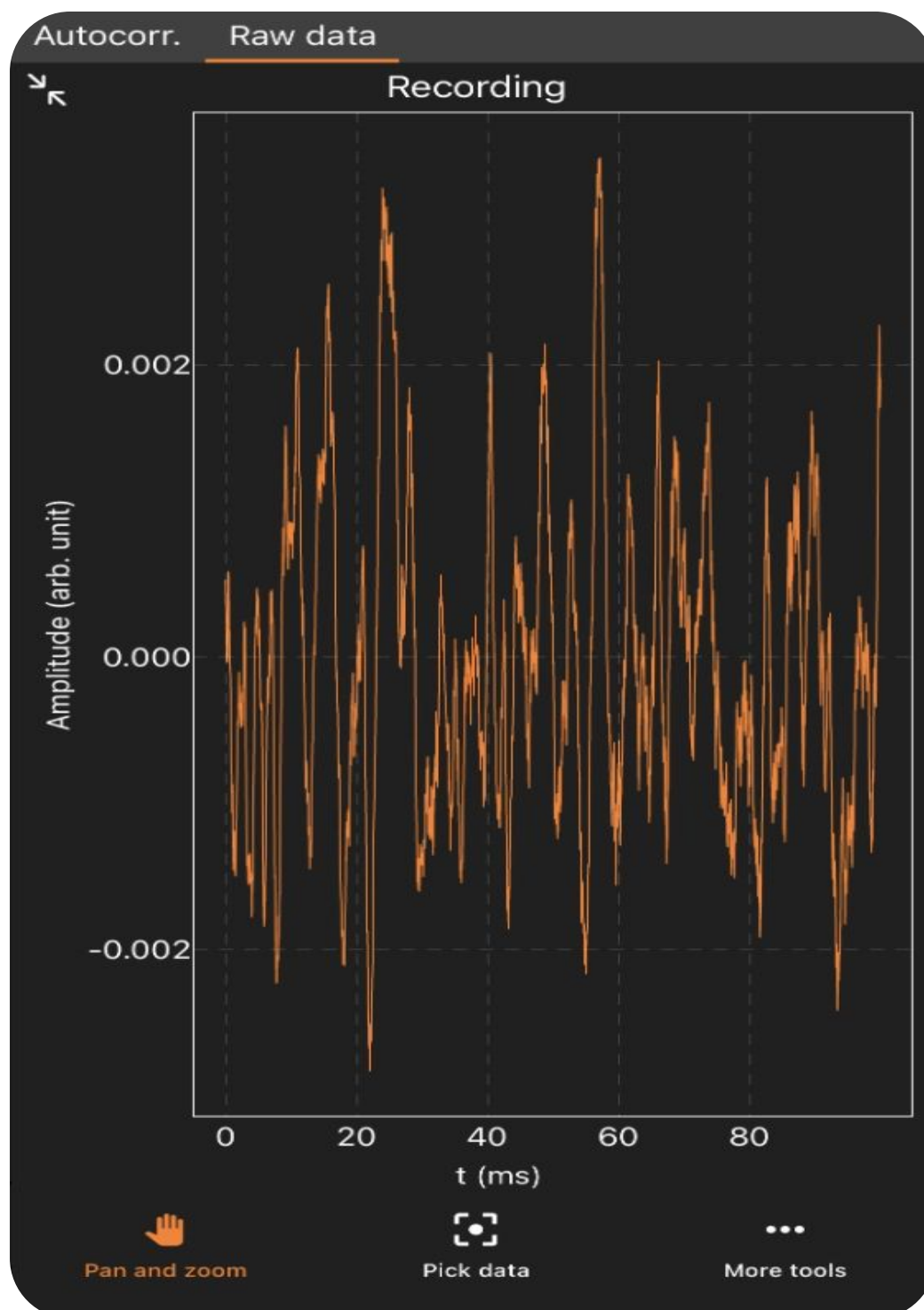
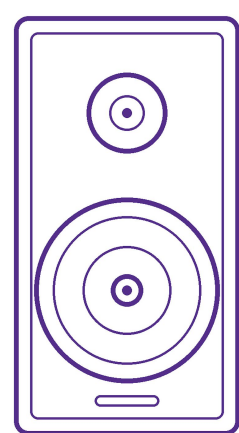


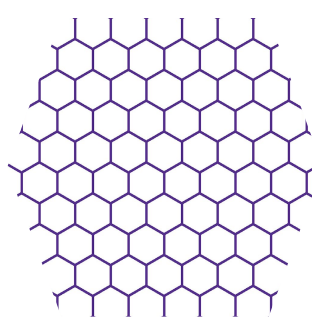
Fig. 6: The frequency data from PhyPhox

7. SEEKING HELP

We sought help from Professor V. Hunter Adams, of Cornell's Electrical and Computing Engineering they were happy to lend us their ear. From them we learnt:



It's better to use a speaker to achieve frequency in the 10's of Hz. However, due to our budget restraint we remained with a motor.



The zapping device works better when there are two sheets of chicken wire, and the SLFs legs are the conducting connection, as all the electricity flows through the SLF, killing it. As shown below

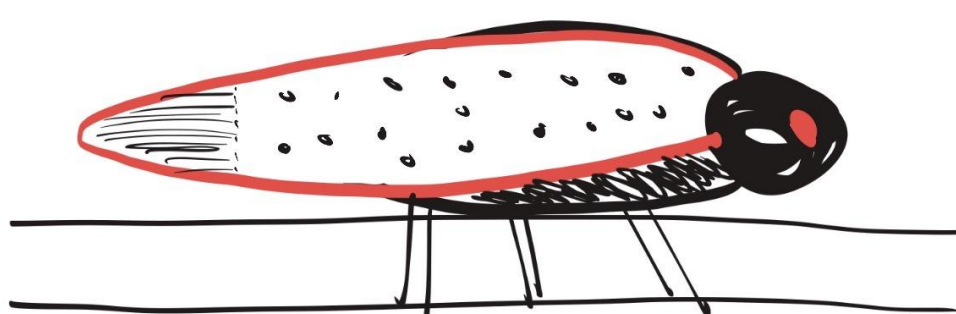


Fig. 7: An SLF perching on one sheet of chicken wire. Some legs reach through to the other side, causing the conduction of electricity and killing of the SLF

8. FINAL DESIGN

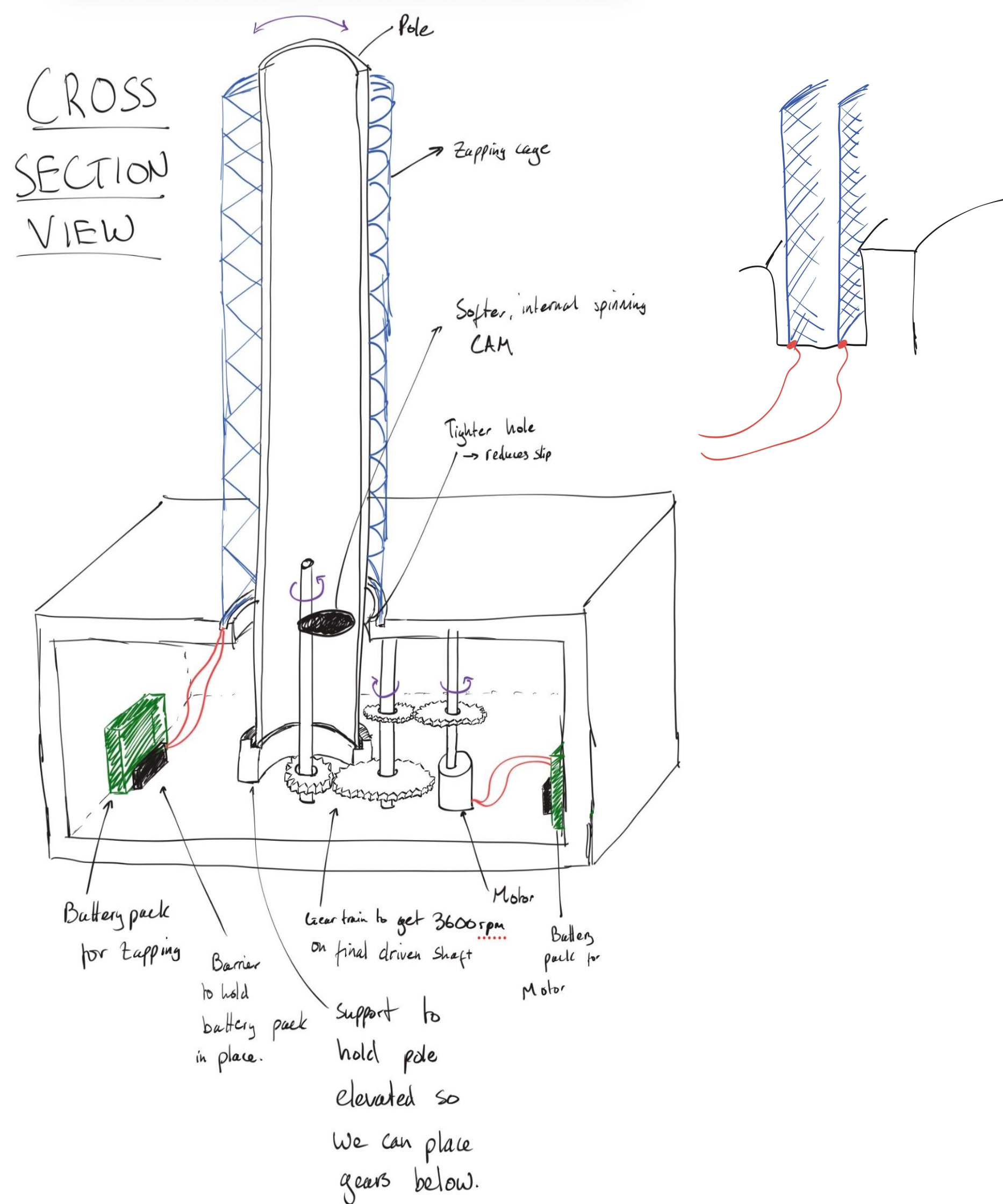


Fig. 8: Our Final sketch plus an excerpt of the zapping system in the top right hand corner. On the gear train left to right we have 12 teeth, 32 teeth & 14 teeth, 14 teeth which give us 3600 rpm on the final driven shaft.

The final sketch includes the new gear train, double sheet zapping system, smooth rubber CAM, and raised pole design.

Through research, trial and error, and consultations with experienced individuals we are happy with this final design.

9. FINAL TESTING

Having made improvements to our model, we needed to re-test its capabilities to ensure it met our success criteria

- Using PhyPhox again we found that the frequency was exactly 60 Hz
- After 5 minutes of continuous running, the mechanism was still in place and running. There was no wear and tear so we are happy that we improved from the first prototype
- Once again, we applied a lateral force to the pole. This time, 10N were reached before deformation, proving our adjustments to the model were successful.

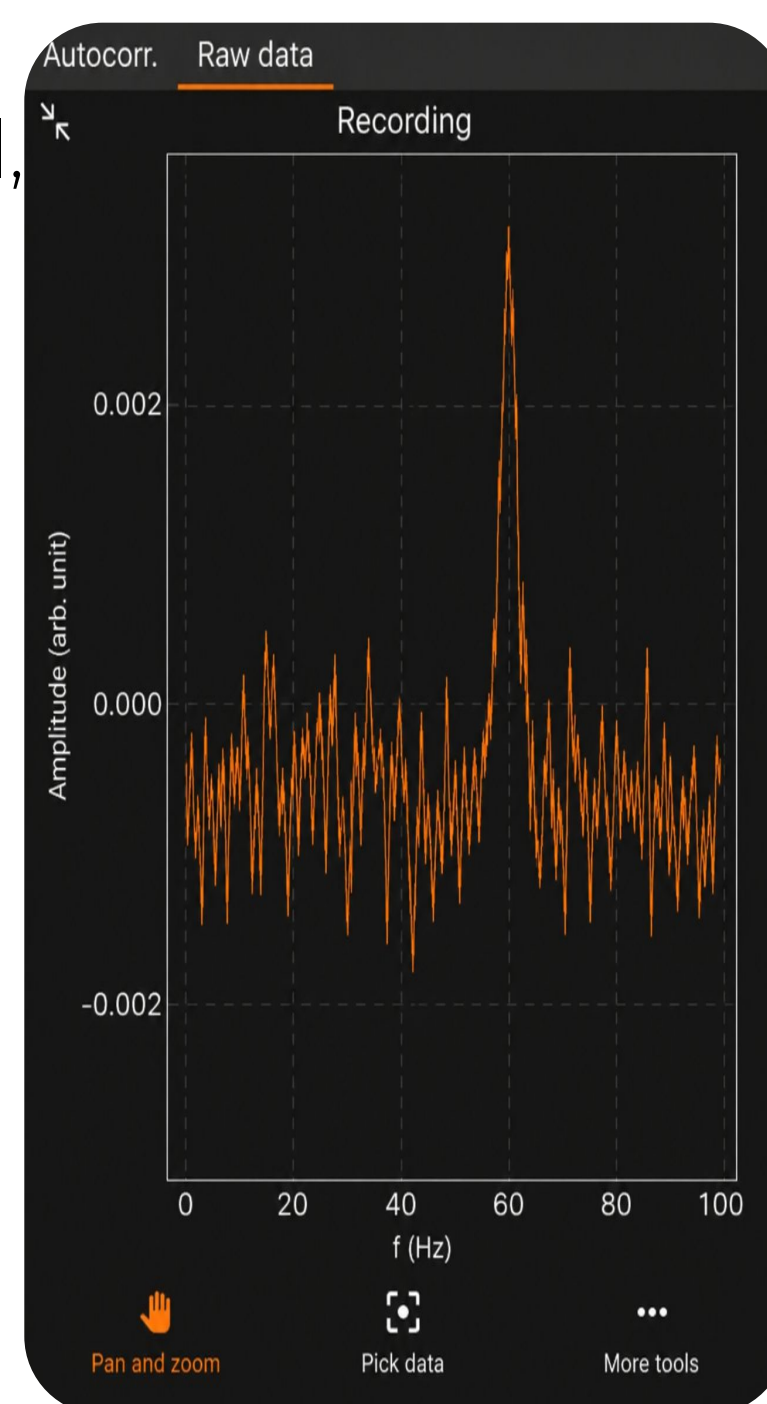


Fig. 9: The frequency data from PhyPhox for the final test

10. CONCLUSION AND RECOMMENDATIONS

Was our design successful?

- Yes! It met all of our success criteria and we are happy with its capabilities

What would we change if we got taken to market? (more materials, time, larger budget)

- Make everything out of a weatherproof material (i.e. stainless steel)
- Use a speaker instead of a motor
- Scale the design significantly so same size as a tree
- Laser engrave the pole so it has texture so the SLFs can latch onto it